

Cat Toy Prototyping Process

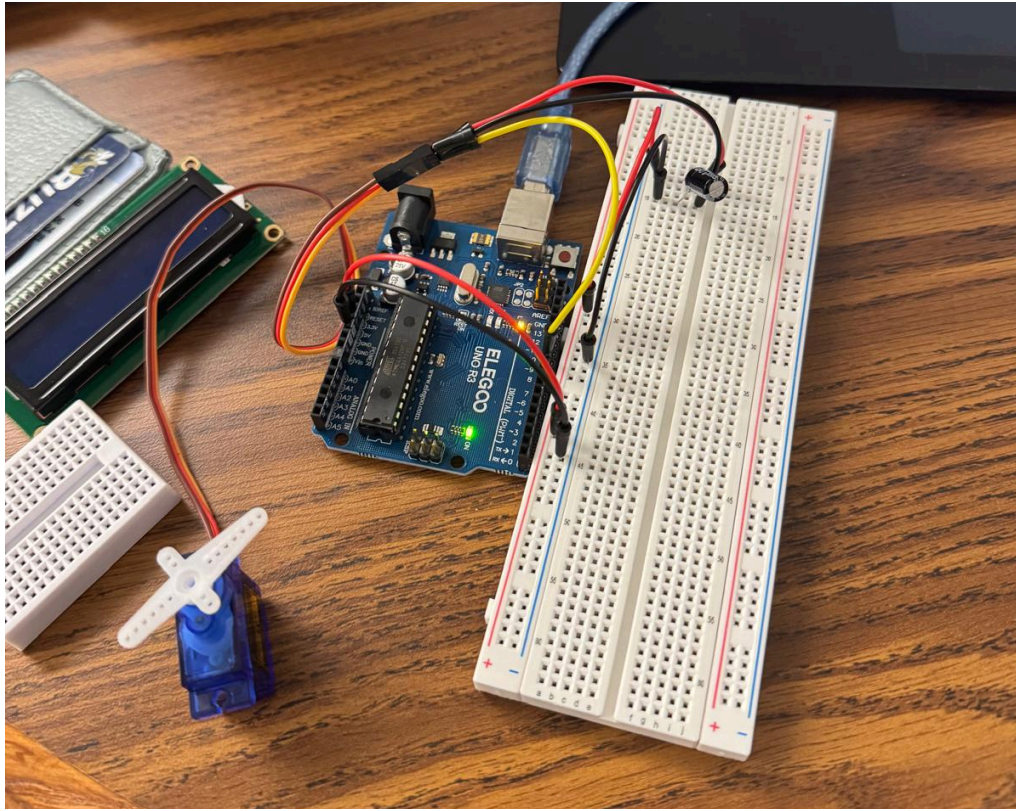


Figure 1. Set-up Servo motor.

- Tested servo motor on simple arduino code
- Servo motor rotate and sweep functions both worked.

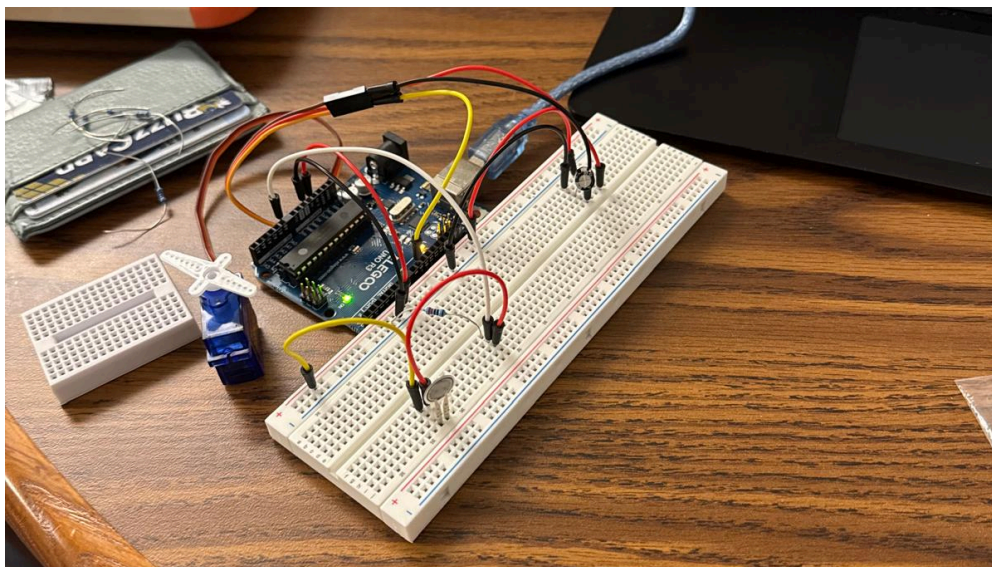


Figure 2. Added FSR.

- FSR inputs took too long and were sometimes finicky
- This likely wouldn't work with cats, since they move very quickly - too fast for the FSR to register
- Considered switching to a photoresistor, but after some more research, photoresistors are about as slow as FSRs.
- PIR sensor seemed like the best option.
- Integrated PIR sensor into circuit

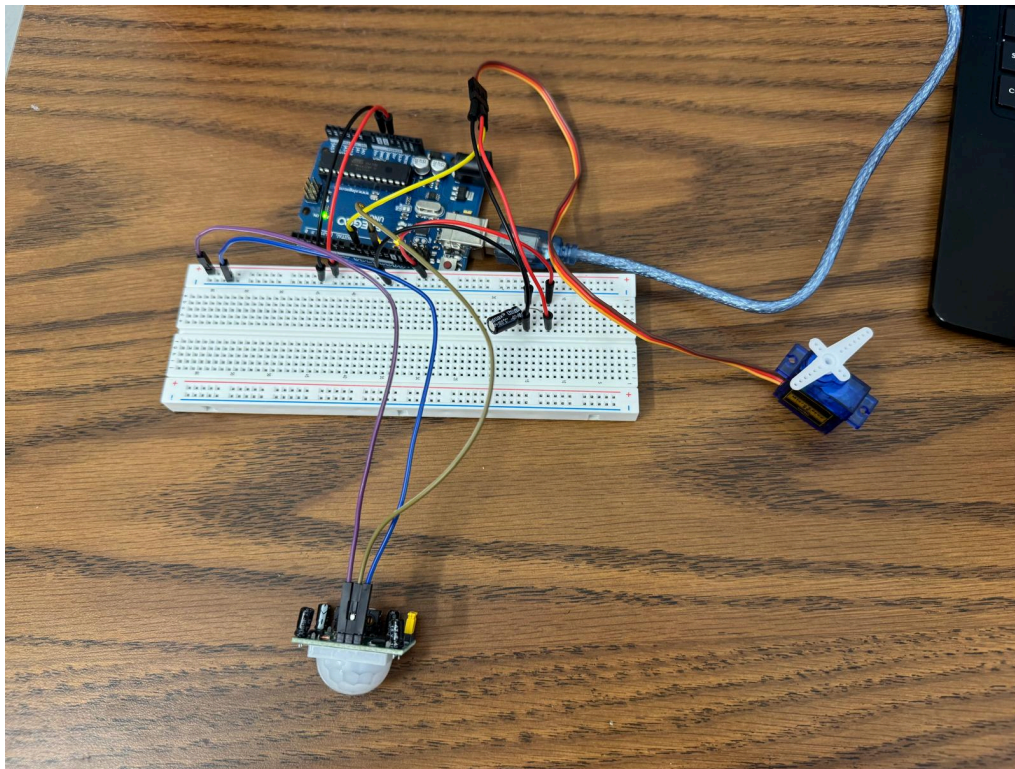


Figure 3. Final circuit with PIR sensor + servo motor.

- The PIR sensor originally wasn't detecting any movement.
- Double checked variable time delay and adjusted to the shortest possible time.
- Wrote initial code for servo motor to respond to PIR sensor without state dependency.
- Added state logic, but the PIR sensor stayed stuck at the ENGAGE state.
- Debugged code to treat motion as an event rather than a continuous state, since the PIR can stay high for a longer period of time due to time delays.

Skills Gained:

- Embedded systems thinking
- Closed-loop systems (how does feedback change behavior?)
- Finite state machines
- Hardware-software integration
- Debugging
- Behavioral Design